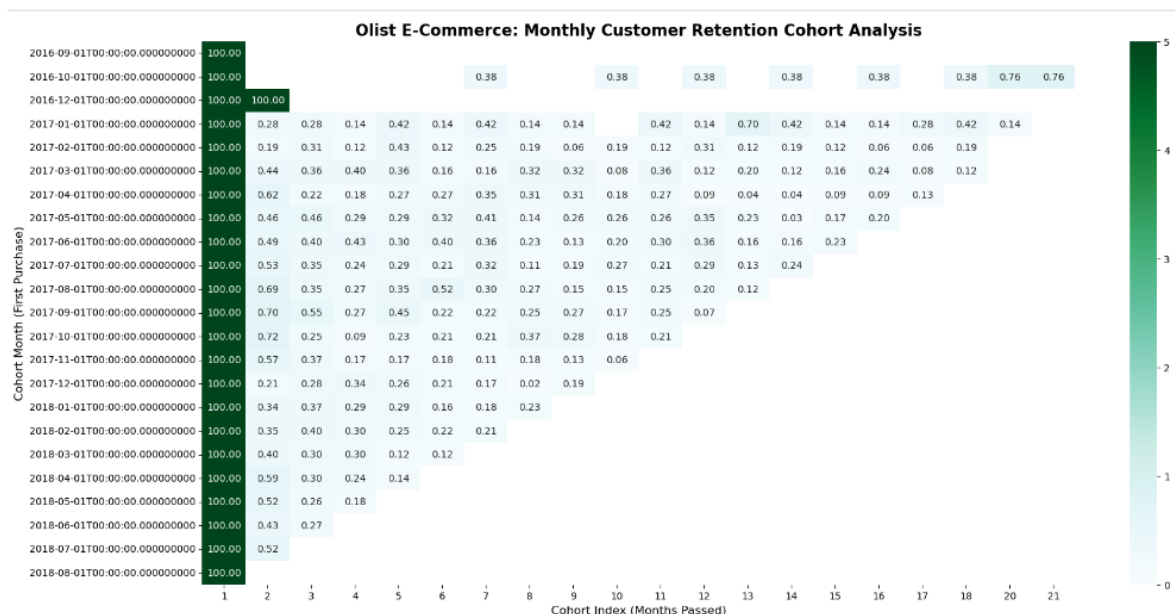




PROJECT REPORT: Olist E-Commerce - Customer Lifecycle & Cohort Analytics

3. Key Performance Indicators (KPIs)

The executive view displays 3 primary high-level business metrics that reflect Olist's overall health:

1. **Total Revenue: \$13,218,400** — Indicates the aggregate gross sales volume processed across the platform.
2. **Total Customers: 93,336 Unique Users** — Measures the scale of Olist's customer acquisition and active user base.
3. **Avg. Order Frequency: 1.033** — Reveals that, on average, the majority of the customer base has only made a single purchase, highlighting a critical operational bottleneck.

4. Advanced Analytics & Core Components

A. RFM Customer Segmentation

Customers are dynamically categorized based on how recently they bought, how often they buy, and how much they spend:

- **VIP / Champions:** High-value customers who buy frequently, spend generously, and have purchased recently.
- **Loyal Customers:** Consistent buyers who provide stable revenue streams and steady engagement.
- **New Customers:** Recently acquired users with high potential who require onboarding focus.
- **Lost Customers / At Risk:** Previously active buyers who have drifted away and have a high probability of churn.

B. Cohort Retention Heatmap

- **Cohort Index (Months 1 to 18):** Tracks monthly customer groups from their first purchase date to monitor recurring activity over time.
- **Key Insight:** While initial acquisition (Cohort Index 1) is robust, the subsequent drop-off is steep. The low retention rate in later months proves that Olist faces a major structural challenge in retaining users, signaling that shifting focus from acquisition to customer lifetime value (LTV) is urgent.

5. Dashboard Engineering & Interactivity

- **Data Blending & Custom Relationships:** Formed a dynamic backend schema connecting two separate datasets (cleaned_olist_cohort_data and olist_rfm_segments.csv) using Customer Unique Id as the primary key.
- **Cross-Filtering Actions (Use as Filter):** The entire system is built to be fully interactive. Selecting any customer tier in the RFM bar chart automatically filters both the top-level KPIs and the Cohort Heatmap instantly to reflect that specific segment's performance.

6. Strategic Business Recommendations

1. **Implement Retention Programs:** Given the low average order frequency of 1.033, Olist must introduce post-purchase engagement workflows, automated

PROJECT REPORT: Olist E-Commerce - Customer Lifecycle & Cohort Analytics

email marketing loops, and a structured loyalty points system to incentivize a second purchase.

- 2. **Win-Back Campaigns:** Deploy targeted win-back strategies with exclusive discount codes, personalized recommendations, or feedback surveys

